

Levosimendan Treatment Model for Post-Ischemic Stroke

BIOE3001 Quantitative Methods in Biomedical Engineering

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Introduction

Ischemic stroke occurs when an artery that supplies the brain with nutrients is obstructed, starving it of its constant supply from the heart. According to the World Stroke Organisation, it is the second leading cause of death worldwide, with 11.9 million new strokes and 7.52 million deaths occurring each year (Feigin et al., 2025). In Australia, about 425,000 people are living post-stroke and around 11% of patients have a subsequent stroke within a year (AIHW, 2026) (Peng et al., 2025). The brain receives about 15% of cardiac output and requires a continuous supply of about 50mL/100g/min of blood (MJ, 2009). Cerebral autoregulation maintains a constant flow across mean arterial pressures (MAP) of about 50-150mmHg. Arterioles constrict when perfusion pressure rises and dilate when it falls. Experiencing an ischemic stroke not only impairs this biological control but also collapses the pressure arriving at the cerebral tissue. This, in consequence, significantly hinders the blood flow arriving at the cerebral vasculature and falls below the homeostasis threshold level of survival for brain tissue (Pires et al., 2013). Some cohorts recover within several months; however, according to the patient database supplied for this project, chronic stroke cohorts could only recover 70% of MCA velocity following a stroke in comparison to the age-matched control cohort (Novak et al., 2010). Cerebral flow depends on the heart's output pressure and pump rate and the resistance of the cerebral bed.

Current treatment involves reopening the blocked cerebral artery. Thrombolysis rarely clears clots longer than about 8mm (Riedel et al., 2011) and thrombectomy is limited to large vessels within a feasible time window (Goyal et al., 2016). Neither restores the biological control for vascular autoregulation (Nam et al., 2023). Levosimendan is an Inodilator used in acute heart failure but has been shown to increase MCA velocity (Follath et al., 2002). It raises stroke volume by sensitising troponin C to calcium and lowers resistance by opening vascular K+ channels (Papp et al., 2012). The drug itself clears within hours (0.9h); however, its active metabolite, OR-1896, remains in the blood circulatory system for several days (Puttonen et al., 2008).

This report explores the use of modelling pharmacokinetic and pharmacodynamic systems to optimise the administration and dosing of Levosimendan to restore cerebral blood flow to a healthy, homeostatic level. Using patient cardiovascular parameters, the model will predict Levosimendan behaviour and its repeated behaviour to restore cerebral blood flow velocity toward a matched no-stroke patient whilst keeping heart rate and blood pressure within safe cardiotoxic limits.

Model Formulation

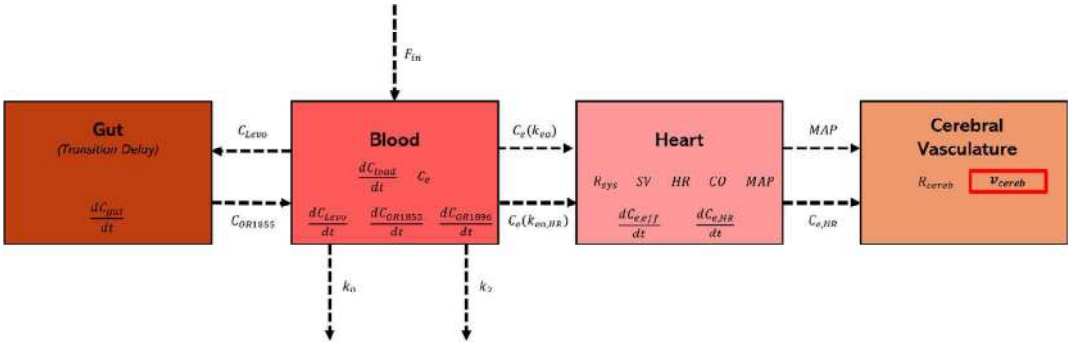


Figure 1: Model Mass Transfer Diagram

Three key physiological systems are involved: The cardiovascular system, which sets cardiac output and arterial pressure; the cerebrovascular system, whose resistance determines how much pressure becomes cerebral; and the circulatory system, where the drug is administered, reduced and acetylated into the long-life active metabolite OR-1896 (Antila et al., 2004). Figure 1: Model Mass Transfer Diagram above represents these systems as mass-transfer blocks, where the blood and gut hold the five pharmacokinetic differential variables, the heart which holds the cardiovascular pharmacodynamic parameters and the cerebral vasculature where the model output (vMCA) is calculated.

Below in Table 1: Equations for modelling drug concentrations, cardio and cerebral dynamics describes these dynamics. There are seven differential equations along with nine algebraic relations that link the effect-site concentrations to the cardiac and cerebral outputs. Together, these equations form the model used to predict cerebral blood flow from Levosimendan dosing.

Table 1: Equations for modelling drug concentrations, cardio and cerebral dynamics

	Description	Equation	Units
		Pharmacokinetics	
(1)	Loading dose (parent)	$\frac{dC_{load}}{dt} = -k_0\, C_{load}$	$\frac{ng}{mL * h}$
(2)	Blood Levosimendan concentration	$\frac{dC_{Levo}}{dt} = \begin{cases} F_{in} - L_{inf} - k_0\, C_{load} & ,\text{infusing} \\ -k_0\, C_{Levo} & ,\text{else} \end{cases}$	$\frac{ng}{mL * h}$
(3)	Formation inside gut	$Form = \begin{cases} m_1 V_{ratio}\left(L_{inf} + k_0 C_{load}\right) & ,\text{infusing} \\ m_1 V_{ratio}\, k_0\, C_{Levo} & ,\text{else} \end{cases}$	$\frac{ng}{mL * h}$
(4)	Transition time release delay from gut	$\frac{dC_{gut}}{dt} = \text{Form} - \text{Rel}, \quad \text{Rel} = \begin{cases} 0 & ,t < T_{gut} \\ k_g\, C_{gut} & ,t \geq T_{gut} \end{cases}$	$\frac{ng}{mL * h}$
(5)	Blood OR-1855 metabolite concentration	$\frac{dC_{OR1855}}{dt} = \text{Rel} - k_{ac}\, C_{OR1855}$	$\frac{ng}{mL * h}$
(6)	Blood OR-1896 metabolite concentration	$\frac{dC_{OR1896}}{dt} = k_{ac}\, C_{OR1855} - k_2\, C_{OR1896}$	$\frac{ng}{mL * h}$
(7)	Free concentration of active drug	$C_e = f_0\, C_{Levo} + f_2\, C_{OR1896}$	$\frac{ng}{mL}$
		Pharmacodynamics	
(8)	Slow-effect site (Stroke volume)	$\frac{dC_{e,eff}}{dt} = k_{eo}\left(C_e - C_{e,eff}\right)$	$\frac{ng}{mL * h}$
(9)	Fast-effect site (HR and vessels)	$\frac{dC_{e,HR}}{dt} = k_{eo,HR}\left(k_{HR}\, f_0\, C_{Levo} + f_2\, C_{OR1896} - C_{e,HR}\right)$	$\frac{ng}{mL * h}$
(10)	Stroke volume	$SV = SV_0\left[1 + \frac{dSV_{max}\, C_{e,eff}}{EC_{50} + C_{e,eff}}\right]$	m^3
(11)	Heart rate	$HR = HR_0\left[1 + \frac{E_{max,HR}\, C_{e,HR}}{EC_{50} + C_{e,HR}}\right]$	Hz
(12)	Systemic resistance	$R_{sys} = R_{sys,0}\left[1 - \frac{E_{max,R}\, C_{e,HR}}{EC_{50} + C_{e,HR}}\right]$	$\frac{Pa * s}{m^3}$
(13)	Cardiac output	$CO = SV \cdot HR$	$\frac{m^3}{s}$
(14)	Mean arterial pressure	$MAP = CO \cdot R_{sys}$	Pa
(15)	Cerebral resistance	$R_{cereb} = R_{cereb,0}\left[1 - \frac{E_{max,R}\, C_{e,HR}}{EC_{50} + C_{e,HR}}\right]$	$\frac{Pa * s}{m^3}$
(16)	Cerebral blood velocity	$v_{cereb} = \frac{MAP - P_{brain}}{R_{cereb}\, A_{cereb}}$	$\frac{m}{s}$

Pharmacokinetics:

Equation (1) tracks the loading dose as a separate part of the parent drug dosing, which decays at the parent elimination rate k_0. Equation (2) describes the blood concentration of the parent drug Levosimendan over time. During infusion, input F_{in} is opposed by a constant absolute loss which linearly scales against the F_{in} dose quantity. This input dynamic was modelled by the dynamics in Table as it linearly rises to 24h instead of a flat plateau during the 24hr constant. This loss term was derived in Appendix A: Supplementary Equations and Parameters for Model After the infusion, parent drug decays out with first-order half-life of 0.9h (Puttonen et al., 2008). Approximately 5% of the parent drug is reduced in the gut to the intermediate OR-1855. The remaining drug compound is conjugated with inactive compounds and leave through urine and faeces (Antila et al., 2004). In Model Implementation, this loss is modelled by L_{inf}.

Equation (3) models the transfer of Levosimendan from the blood to the gut, dictated by the fractional amount m_1. Then, V_{ratio} converts this from the parent's volume of distribution to the metabolite's. This term was fitted from Table 1 data. Since the metabolites (both OR-1855 and 1896) don't appear in blood plasma immediately after infusion, Equation (4) represents the time delay of the OR-1855 metabolite as a holding compartment. The metabolite release Rel is toggled off until the time passed has passed T_{gut}. Without this term, metabolite concentrations would have peaked early, and it would not match Table 1 data.

Equation (5) represents the OR-1855 intermediate metabolite. It has no pharmacodynamic effect and is used as an intermediate for OR-1896, the active metabolite. OR-1855 accumulates in blood plasma from the release from the gut defined in Equation 4 and then its lost to N-acetylation by the enzyme NAT2 to form OR-1896. There was originally a decay term for OR-1855, however when fitting to Table 1 data, it was found that acetylation alone accounts for majority of OR-1855 loss. It was thus removed to reduce a term from Equation 5 and one parameter for fitting. Equation (6) represents the final active metabolite OR-1896. It is formed by the acetylation term in Equation (5) and eliminated through first-order decay k_2. Pharmacologically wise, the acetylation reaction is reversible; however, that term was removed as it only affected the cerebral blood velocity output by 2%. Finally, Equation (7) models the free concentration of active drug C_e available to act on the heart and cerebral blood vessels. Only unbound drug is pharmacologically active so parent and OR-1896 concentrations are each weighted by free fractions f_0 and f_2 respectively. Levosimendan is 97% protein-bound, whereas OR-1896 is only 40% bound (Antila et al., 2004) Therefore, OR-1896 dominates after 24 hours, consistent with the haemodynamic effects lasting for several days after a single 24 hour infusion (Lilleberg et al., 2007).

Pharmacodynamics:

Equation (8) models the slow effect site C_e, eff representing the delay between plasma concentration and the heart contraction. The effect lags plasma concentration so C_e, eff follows C_e with first order decay k_{eo}. Equation (9) models a second, fast effect site $C_{e,HR}$ for heart rate and vascular resistance, that weights the parent drug more. Two drug-effect terms are needed because in Table 3, heart rate and MAP respond within 6h with parent drug however, volume rises more slowly over 12h with OR-1896.

Equations (10) to (12) model the drug's direct effect on stroke volume, heart rate and systemic resistance using Emax (Hill) kinetics. Each response increases with concentration and saturates at its maximum. Stroke volume and heart rate rise and systemic resistance falls, all by their E_{max} terms. All three share one EC_{50}, since separately fitted values converged to the same value after parameterisation.

Equation (13) represents cardiac output as the product of stroke volume and heart rate. Equation (14) calculates MAP from derivation of Ohm's law (Robinson & Allenby, 2026). The baseline resistances $R_{sys,0}$ and $R_{cereb,0}$ are back calculated from the patient's own values. See Appendix A: Supplementary Equations and Parameters for Model for full details. Equation (15) models cerebral resistance with the same Emax term as systemic resistance, as cerebral and systemic vessels dilate through the same mechanism (Yildiz, 2007). Finally, Equation (16) calculates the model output v_{cereb} from the pressure drop across a lumped cerebral resistance. This replaced the original extended-Bernoulli formulation in Appendix A: Supplementary Equations and Parameters for Model.

Parameter Estimation

Table 2: Initial parameter estimations for model

Symbol	Description	Initial value	Units	Source
		Pharmacokinetics		
$t_{1/2,0}$	Levosimendan half-life	0.9	h	Puttonen et al. (2008)
k_0	Levosimendan elimination rate	0.77	$\frac{1}{h}$	Puttonen et al. (2008)
C_{slope}	Net parent rise during infusion	1.3388	$\frac{ng}{mL * h}$	Puttonen et al. (2008)
$C_{0,levo}$	Distribution jump at start	25.0	$\frac{ng}{mL}$	Puttonen et al. (2008)
V_{kg}	Parent volume of distribution	132	$\frac{mL}{kg}$	Estimated from Puttonen et al. (2008)
T_{gut}	Gut release onset	30	h	Puttonen et al. (2008)
k_g	Gut release rate	0.0131	$\frac{1}{h}$	Bertin et al. (2026)
m_1	Fraction reduced to OR-1855	0.05	–	Antila et al. (2004)
V_{ratio}	Parent-to-metabolite volume ratio	0.5	–	Estimate
$t_{1/2,ac}$	NAT2 acetylation rate	9.6	h	Bertin et al. (2026)
$t_{1/2,2}$	OR-1896 elimination half-life	7.7	h	Estimate
k_2	OR-1896 elimination rate	0.09	$\frac{1}{h}$	Puttonen et al. (2008)
f_0	Free fraction, Levosimendan	0.03	–	Antila et al. (2004)
f_2	Free fraction, OR-1896	0.61	–	Puttonen et al. (2008); Antila et al. (2004)
		Pharmacodynamics		
$t_{1/2,eo}$	Slow effect delay (SV)	10	h	Estimate
$t_{1/2,eo,HR}$	Fast effect delay (HR, vessels)	6	h	Álvarez et al. (2006); Lilleberg et al. (2007)
k_{HR}	Parent potency weight, fast site	6	–	Álvarez et al. (2006)
EC_{50}	Shared half-maximal concentration	0.5	$\frac{ng}{mL}$	Estimate
dSV_{max}	Maximum fractional SV rise	0.28	–	Slawsky et al. (2000)
$E_{max,HR}$	Maximum fractional HR rise	0.11	–	Kivikko et al. (2015); Álvarez et al. (2006)
$E_{max,R}$	Maximum fractional fall in R_{sys} and R_{cereb}	0.40	–	Estimate

Table 2: Initial parameter estimations for model above lists the initial parameter values used in the model (Code #1). Most are taken from literature or read directly from Table 1 data. The rest are reasoned estimates that were calibrated from Code #1 model development since there are no published values exist. Half-lives were converted to constants $k = ln(\frac{2}{t_{1/2}})$ and patient baselines and dosing parameters are in Appendix A: Supplementary Equations and Parameters for Model. All differential variables start at 0ng/mL except C_{Levo}. Model is simulated from 0 to 288 hours (12 days). Estimated values are refined in Model Parameterisation.

Model Implementation

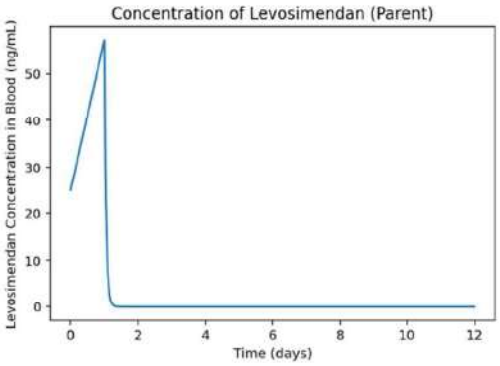


Figure 2: Levosimendan blood concentration over time

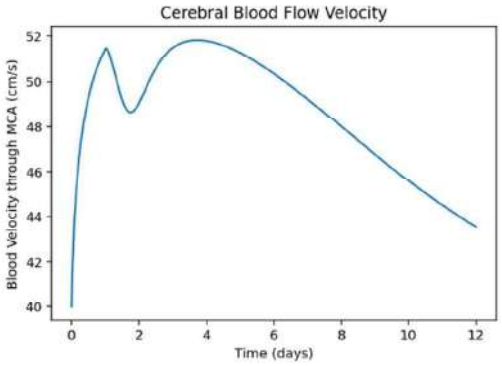


Figure 3: Cerebral blood velocity (vMCA) over time

The model code was implemented such that the user can customize the dosing regimen and scheduling in the code. See Appendix 1: Code 1 (Model Implementation). This flexibility of dosing customisation was used to allow the model to simulate different dosing regimens such as the Table 3 dosing regimen and is used during Model Parameterisation such that the model robust enough to account for different dosing regimens. The equations in Table 1Table 1 were implemented using the Table 1 dosing parameters for an 80kg patient (Puttonen et al., 2008) This infusion follows the dosing guidelines from Simdax Levosimendan guidelines of 0.1ug/kg/min for an 80kg individual (*Simdax (Levosimendan)*, 2025). Infusion is switched on while $T_{rep} < T_{inf}$ and gut release is held off until T_{gut} = 30h. The model's single infusion mode runs for 12 days, which correlates to the drug's effective time (insert) and allows comparison with Table 1 to capture and compare PK dosing and metabolite generation and elimination. The repeat dosing dosing mode runs for 30 days, restarting infusion every T_{rep} days to show how active drug carries over between doses. In Figure 1Figure 2, Levosimendan jumps to 25ng/mL, rises linearly to 57ng/mL at 24hrs and decays out 1½ days once infusion stops. This matches the kinetic data captured in Table 1. In Figure 3, v_{cereb} rises to 51.4cm/s by 24h, which is driven by Levosimendan raising heart rate and MAP (Figure 16 and Figure 17 in Appendix B: Model Outputs for implementation. Although parent drug washes out, it rises again and reaches a second peak of 51.8cm/s at roughly 4 days after infusion. This is driven by the active metabolite OR-1896, which is consistent with the multi-day effect stated in (Lilleberg et al., 2007). Further details of PK PD parameter output behaviour are in Appendix B.

Model Parameterisation

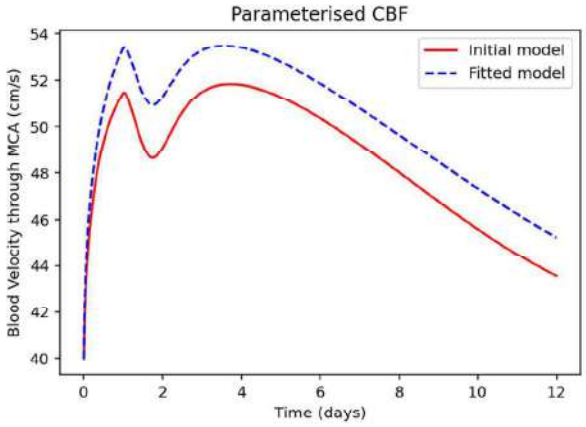


Figure 4: Parameterised Model Output (vMCA)

Fitted Model Parameters

Parameter	Initial	Fitted	SE	Change	Units	Set by
gut_rate	53	65.9	7.02	24.3%	hrs	Table 1
Vm_ratio	0.5	0.5275	0.0671	5.5%	-	Table 1
ac_rate	9.6	4.951	0.385	-48.4%	hrs	Table 1
m2_half	7.7	7.642	0.758	-0.8%	hrs	Table 1
f0	0.03	0.03	-	0.0%	-	Fixed
keo_thalf	10	13.85	4.34	38.5%	hrs	Table 3
Emax_R	0.4	0.3675	0.00713	-8.1%	-	Table 3
EC50	0.5	0.5	-	0.0%	ng/mL	Fixed
dSVmax	0.28	0.28	-	0.0%	-	Fixed
Emax_HR	0.11	0.153	0.00522	39.1%	-	Table 3
k_HR	6	6	-	0.0%	-	Fixed
keo_HR_thalf	6	6	-	0.0%	hrs	Fixed

Fit Quality (initial -> fitted)

Output	N	R2 init	R2 fit	RMSE init	RMSE fit	Units
T1 Levosimendan	21	0.992	0.992	1.536	1.536	ng/mL
T1 OR-1855	21	-0.171	0.905	0.801	0.228	ng/mL
T1 OR-1896	21	0.908	0.972	0.265	0.145	ng/mL
T3 HR	6	-4.983	0.645	3.441	0.838	bpm
T3 MAP	6	-7.792	0.818	5.575	0.801	mmHg

Figure 5: Fitted parameters and parameter fit metrics

The model was then further parameterized to fit both Table 1 and Table 3 datasets to capture how the drug affects cardiac and cerebral dynamics across different dosing regimens. Stage 1 of the process fitted four PK parameters to the Table 1 data. Stage 2 then held fixed PK values (after parameterising with Table 1) and fitted three PD parameters to the Table 3 HR and MAP data under its own regimen (Álvarez et al., 2006). Both stages used bounded least squares to minimise residual errors. With all PD parameters free, the fit was unidentifiable, so f_0 and dSV_{max} were fixed to literature values. The fast effect site was also added because a coupled drug-effect site could not match both the fast HR/MAP response and the slower stroke volume dynamics in Table 3 data. In Appendix C: Parameterisation additional material, Figure 18 outlines the process for Parameterization, whilst Figure 19 and Figure 20 compare the estimated and model-fitted plots with the Table 1 (PK) and Table 3 (PD) data whilst maintaining their own separate dosing regimens. After running the parameterization fitting, OR-1855 R^2 from −0.17 to 0.91, OR-1896 from 0.91 to 0.97, HR from −4.98 to 0.65 and MAP from −7.79 to 0.82, from Figure 5. The parent was unchanged, as it is set by fixed parameters. The fitted v_{cereb} peaks at 53.5 cm/s and stays above 50 cm/s for roughly 8 days Figure 4. Table 3 parameters couldn't be fitted perfectly because Table 3 only has seven points over 48 h, limiting the range of data to work with.

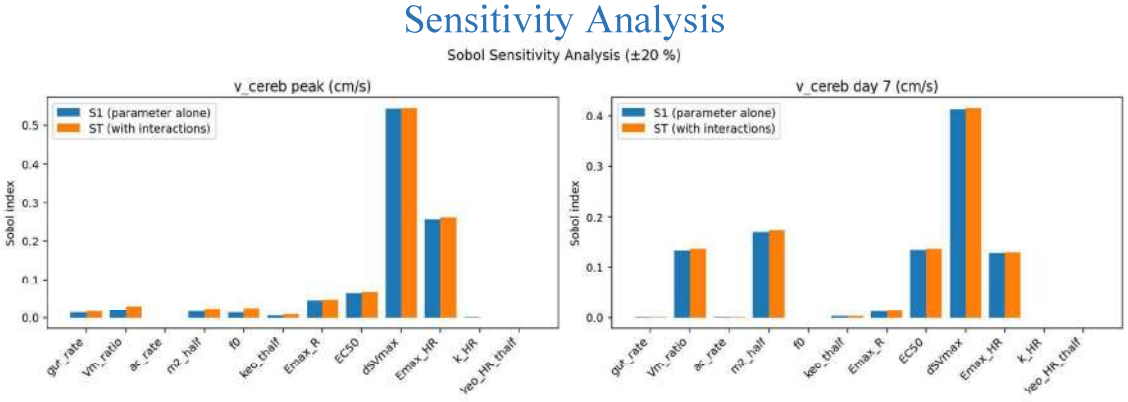


Figure 6: Sensitivity parameter-driven analysis of model outputs at peak and effective time end

A Sobol sensitivity analysis was performed on the parameterised model to identify which parameters drive v_{cereb} at its peak and when it reaches 50cm/s, the range of no-stroke controls in the Novak et al. patient dataset. The implementation was based off the Sobol analysis from (Mann & Allenby, 2026) and the process is outlined in Figure 27 in Appendix D: Sobol Sensitivity Analysis additional material. Figure 6 and Table 3 show that velocity is mainly driven by the cardiac output: dSV_{max} (S_1 = 0.55) and

$E_{max,HR}$ (0.26) explain drive 80 % of the peak. By day 7, the parameters that control OR-1896 also matter: $t_{1/2,2}$, V_{ratio} and EC_{50} (0.13–0.17). Interactions are negligible, since $S_T \approx S_1$ and fixed parameters have $S_T \leq 0.03$, which justifies fixing them (Figure 28 in Appendix D). The most influential parameter, dSV_{max}, is a fixed heart-failure value from Slawsky et al., (2000). Only 67 % of samples stay at or above 50 cm/s at day 7, which supports a repeat interval of less than seven days.

Table 3: Sobol indices of parameters with largest effect on cerebral blood flow

	Parameter	S1	S1 confidence	ST	ST confidence
Peak v_{cereb}	$dSVmax$	0.546	0.059	0.547	0.047
	$Emax_{HR}$	0.258	0.04	0.262	0.022
v_{cereb}, day 7	$dSVmax$	0.413	0.05	0.415	0.035
	$m2_{half}$	0.169	0.035	0.172	0.017
	Vm_{ratio}	0.133	0.028	0.136	0.013
	$EC50$	0.134	0.031	0.136	0.012
	$Emax_{HR}$	0.128	0.03	0.13	0.011

Data Processing and Usage

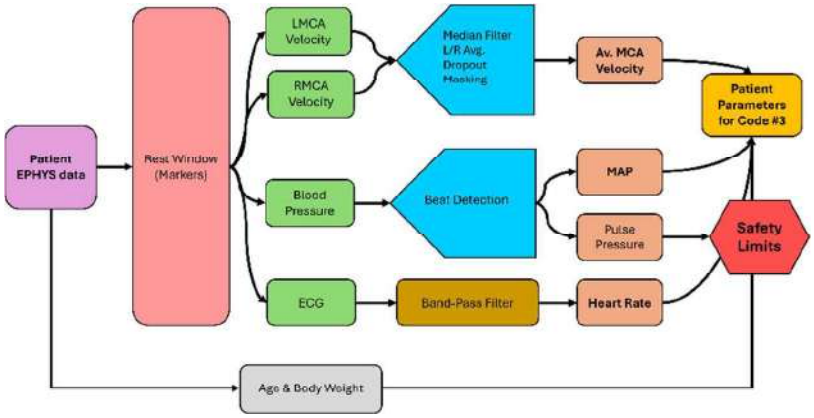


Figure 7: Flowchart for processing EPHYS patient data

Patient data was sourced from Novak et al. (2010). Each stroke patient was paired with an age- and weight-matched no-stroke control as its target: S0175 → S0157 (hypertensive) and S0199 → S0200 (normotensive). In the data processing method shown above in Figure 7,Table 2 only the resting window was used, because the later test phases deliberately change CO_2 and blood pressure. MCA velocity was median-filtered to remove Doppler spikes and beats with velocity ≤ 0 were rejected as signal dropouts. Left and right means were averaged to match the model's single cerebral bed. MAP and pulse pressure were taken from beat-detected blood pressure and HR from R-peaks of the band-pass filtered ECG (Mann & Allenby, 2026). Figure 8 and Figure 9 show S0175 before and after processing, and Figures 29 to 36 in Appendix E: Additional plots for dose-specific patient treatments show all four patients. The resulting HR_0, MAP_0 and $v_{cereb,0}$ in Figure 10 replace the literature baselines in Table 2, and pulse pressure sets each patient's MAP safety floor.

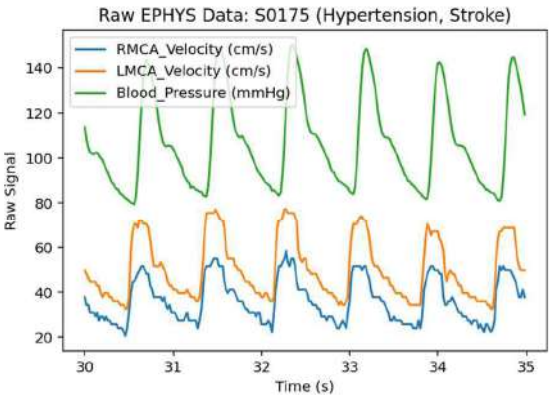


Figure 8: Raw patient PD data for S0175

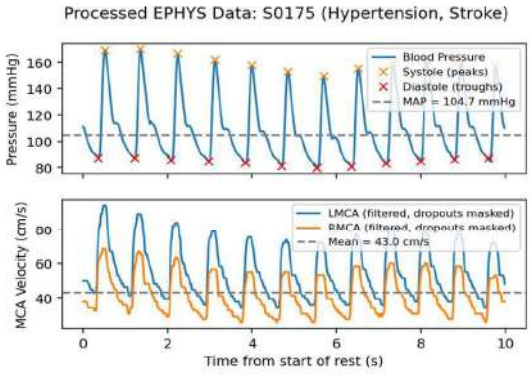


Figure 9: Fully processed patient PD data for S0175

EPHYS Data Processing

Patient	Age (y)	BW (kg)	HR (bpm)	MAP (mmHg)	LMCA (cm/s)	RMCA (cm/s)	Averaged MCA (cm/s)	Dropped Beats (L/R %)
S0175 (Hypertension, Stroke)	64	77	72.0	104.7	49.0	37.0	43.0	0 / 0
S0157 (Hypertension, No Stroke)	67	74	61.9	83.3	51.9	61.0	56.5	0 / 0
S0199 (Normal Tension, Stroke)	77	87	95.4	82.9	22.9	-	22.9	82 / -
S0200 (Normal Tension, No Stroke)	70	76	63.2	68.0	29.5	46.1	37.8	0 / 3

Figure 10: Table summary of all EPHYS processed patients, both groups

S0199 is an outlier with 99 % of right-MCA beats lost to dropout, its left-side velocity of 23 cm/s reflects a collapse in MCA flow that never recovered after the stroke. It is also at its limits before treatment, with HR (95.4 bpm) above the 90 bpm cap and MAP (82.9 mmHg) below its 85.3 mmHg floor calculated in the data processing above.

Personalised Treatment

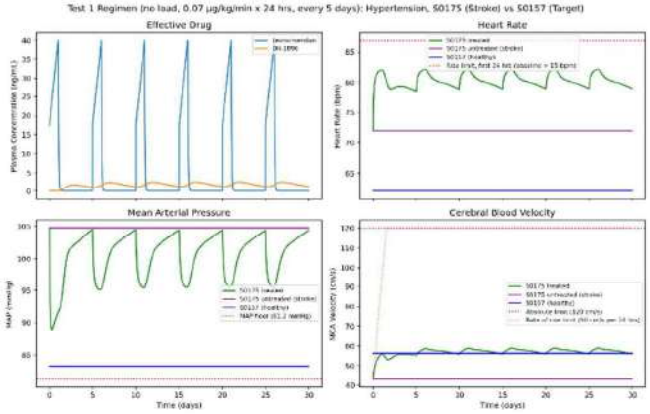


Figure 11: Ideal dosing regimen for hypertension group

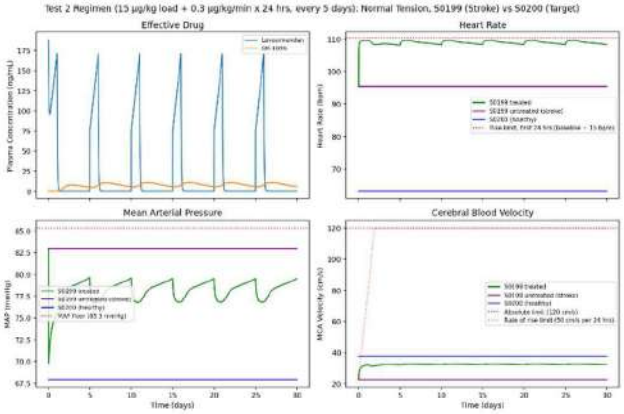


Figure 12: Attempted safe dosing regimen for normal tension group

The aim was to restore each stroke patient's MCA velocity to that of its no-stroke target (S0175 → 57 cm/s, S0199 → 38 cm/s). Each patient's Figure 10 baselines were run for 30 days, with a 24 h infusion every 5 days. The safety limits were a heart-rate rise of no more than 15 bpm in 24 h (Kivikko et al., 2015), MAP at or above 60 + PP/3 (Appendix A), an infusion rate of 0.05–0.2 µg/kg/min (Simdax (Levosimendan), 2025) and velocity at or below 120 cm/s (Lindegaard et al., 1989). The Table 1 and Table 3 regimens tested a standard dose and a higher, loaded dose. Test 1 (0.07 µg/kg/min) was chosen to minimise overshoot in S0175. Test 2 (15 µg/kg + 0.3 µg/kg/min) was a maximum attempt beyond the label limits, made specifically for S0199.

Every regimen brought S0175 to its target safely, as shown in Figures 37 to 44 in Appendix E: Additional plots for dose-specific patient treatments, and Test 1 matched it most closely, peaking at 59.0 cm/s (Figure 11). No regimen restored S0199: even Test 2 peaked at 32.9 cm/s, 87 % of its target, as shown in Figure 12. Velocity can rise only with cardiac output, to at most 1.48 times baseline

(Appendix A), whereas S0199 needs 1.65 times. MAP is also below the safety floor before any drug is administered. Therefore, Levosimendan alone cannot safely restore cerebral flow.

Model and Conclusion

The PK model fits accurately (R^2 0.91–0.99). The PD fit is moderate (HR R^2 = 0.65, MAP R^2 = 0.82), because Table 3 has only seven points over 48 h, taken from a heart-failure cohort. v_cereb is the least certain output: with no time-course data under Levosimendan, it is derived rather than fitted, and it depends mostly on a fixed heart-failure value of dSV_{max}. S0199's baseline is also uncertain, because 82 % of its beats were rejected.

The model has seven ODEs and 19 parameters, seven of them fitted. Structures were added only when the data required them (the gut delay and the second effect site) and removed when negligible (back-conversion, Windkessel compliance, the baroreflex). Sobol analysis shows that five more parameters could be simplified. Sharing EC_{50} and $E_{max,R}$ kept the model identifiable but creates the 1.48× velocity ceiling.

Future work should fit the cerebral arm directly to serial TCD measurements during IV infusion in stroke patients such that the model has cerebral blood flow data to work with. It should also acquire more HR and MAP sampling over 0–6 h and beyond 48 h, and personalise stroke volume and NAT2 acetylator status (Antila et al., 2004). Patients like S0199 need a therapy that lowers cerebral resistance directly rather than acting through cardiac output.

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Appendix

Appendix A: Supplementary Equations and Parameters for Model

Initial Patient Parameters

Used in Parameter Estimation

Symbol	Description	Initial value	Units	Source
HR_0	Baseline heart rate	70	bpm	Böhm et al. (2012)
SV_0	Baseline stroke volume	77	mL	Miller et al. (2021)
MAP_0	Baseline mean arterial pressure	105	$mmHg$	Miller et al. (2021)
$v_{cereb,0}$	Baseline MCA velocity	40	$\frac{cm}{s}$	Miller et al. (2021)
P_{brain}	Downstream (intracranial) pressure	10	$mmHg$	Hood et al. (2023)
r_{cereb}	MCA M1 radius	1.28	mm	Mirza et al. (2024)
$R_{sys,0}$	Baseline systemic resistance	–	–	Derived; see Haemodynamic (PD) derivations
$R_{cereb,0}$	Baseline cerebral resistance	–	–	Derived, see Haemodynamic (PD) derivations

Unit Conversions

Parameter	Clinical Unit	Model Unit	Conversion
Pressure	$mmHg$	Pa	$1mmHg = 133.322\ Pa$
Heart Rate	bpm	Hz	$\div 60$
Stroke Volume	mL	m^3	$1mL = 10^{-6}\ m^3$
MCA Velocity	cm/s	$\frac{m}{s}$	$\div 100$
Time	$days$	h	$* 24$

Dose to plasma input

Infusion distributed into parent volume, then converted to hours:

$$F_{in} = \frac{60 * D_{inf}}{V_{kg}} \left(\frac{ng}{mL * h}\right), C_{load}(0) = \frac{D_{load}}{V_{kg}} \left(\frac{ng}{mL}\right)$$

Used in the model to handle different dosing regimens

Pharmacokinetic (PK) derivations

Levosimendan Infusion: Straight line and constant loss

Used in Model Formulation. Table 1 parent rises linearly during infusion (Puttonen et al., 2008). Implementation became:

$$C_{Levo}(t) = C_{0,levo} + C_{slope}(t), \qquad where: C_{slope} = 1.3388 \left(\frac{ng}{mL * h}\right), C_{slope} \approx 25 \left(\frac{ng}{mL}\right)$$

This fit has $R^2 = 0.94$. Parent jumps to $\approx 25 \left(\frac{ng}{mL}\right)$ before the sample at 2h according to Table 1 data. A constant slope means constant loss:

$$\frac{dC_{Levo}}{dt} = R_{inf} - L_{inf} = C_{slope} \rightarrow \ L_{inf} = (1 - C_{acc}), \qquad where: C_{acc} = \frac{C_{slope}}{60 * \left(\frac{D_{ref}}{V_{kg}}\right)}$$

Only $\approx 3\%$ of infused drug accumulates (C_{acc}). This matches the Table 1 data and rules out first-order loss. C_{acc} is also fixed drug property, measured at $D_{ref} = 100 \left(\frac{ng}{kg}\right)/min$. The distribution scales with dose so that any regimen can be simulated:

$$C_{0,inf} \approx \frac{C_{0,levo}}{D_{ref}} * D_{inf}$$

Metabolite formation: volume ratio

A fraction of m_1 of eliminated Levosimendan concentration becomes OR-1855 metabolite in gut (Antila et al., 2004), (Bertin et al., 2026). Parent and metabolite occupy different volumes, so converting that into metabolite concentrations needs volume ratios:

$$\Delta C_{meta,m1} = m_1 * \left(\frac{V_L}{V_m}\right) * \Delta C_{Levo} = m_1 * V_{ratio} \Delta C_{Levo}$$

From literature, $m_1 = 0.05$ and V_{ratio} is a parameterised term. See Table 2: Initial parameter estimations for model above lists the initial parameter values used in the model (Code #1). Most are taken from literature or read directly from Table 1 data. The rest are reasoned estimates that were calibrated from Code #1 model development since there are no published values exist. Half-lives were converted to constants $k = ln(\frac{2}{t_{1/2}})$ and patient baselines and dosing parameters are in Appendix A: Supplementary Equations and Parameters for Model. All differential variables start at 0ng/mL except C_{Levo}. Model is simulated from 0 to 288 hours (12 days). Estimated values are refined in Model Parameterisation.

Model Implementation

Haemodynamic (PD) derivations

Baseline PD equations

Used in Model Formulation

$$R_{sys,0} = \frac{MAP_0}{SV_0\, HR_0}, \qquad R_{cereb,0} = \frac{MAP_0 - P_{brain}}{v_{cereb,0}\, A_{cereb}}, \qquad A_{cereb} = \pi r_{cereb}^2$$

Initial starting concept: Bernoulli's Principle

The first model implementation treated flow from aortic root to MCA as an energy balance. It added viscous loss term since Bernoulli's principle assumes no friction:

$$MAP + \frac{1}{2}\rho\, v_{ao}^2 = P_{brain} + \frac{1}{2}\rho\, v_{cereb}^2 + R_{cereb}\, A_{cereb}\, v_{cereb}, \qquad v_{ao} = \frac{CO}{A_{ao}}$$

Conceptually:

Aortic Root = MCA Outlet + Pressure Loss from pushing through cerebral vessels

Then solving for v_{cereb}:

$$v_{cereb} = \frac{-R_{cereb}A_{cereb} + \sqrt{(R_{cereb}A_{cereb})^2 + 2\rho\,(MAP - P_{brain}) + \rho^2 v_{ao}^2}}{\rho}$$

Simplification of Bernoulli's Principle

The implementation was simplified for the following reasons:

1. Reduce the parameter count
2. Simplify the computation to a linear, 1st order relationship
3. v_{ao} is velocity of total cardiac output, not just the flow entering MCA
4. Kinetic terms double count losses that a fitted R_{cereb} can account for

The equation was reduced down to a hydraulic Ohm's law across the cerebral bed (Robinson & Allenby, 2026):

$$MAP - P_{brain} = R_{cereb}\, A_{cereb}\, v_{cereb} \rightarrow v_{cereb} = \frac{MAP - P_{brain}}{R_{cereb} * A_{cereb}}$$

MAP Safety Floor

Product information from (*Simdax (Levosimendan)*, 2025) warns against using Levosimendan when diastolic pressure is > 60 mmHg. Model outputs MAP, not diastolic pressure so converting that to MAP is given by:

$$MAP_{floor} = 60 + \frac{PP_0}{3} \text{ mmHg}$$

Patient S0199 already rests below the floor limit before treatment at 82.9.

Velocity Ceiling

Systemic and cerebral resistance share the same drug factor (Hill Kinetics):

$$Drug = 1 - \frac{E_{max,R} C_{e,HR}}{\left(EC_{50} + C_{e,HR}\right)}$$

By substituting this into R_{cereb} and MAP into v_{cereb}, we get this:

$$\frac{v_{cereb}}{v_{cereb,0}} \approx \frac{CO}{CO_0} = \frac{SV}{SV_0} \cdot \frac{HR}{HR_0}$$

Velocity only rises through cardiac output. Cardiac output is bounded and each E_{max} term in Parameter Estimation saturates. So,

$$\frac{CO}{CO_0} \leq (1 + dSV_{max})\left(1 + E_{max,HR}\right) = 1.48$$

This means it can only have a maximum rise of 48%. Due to S0199's collapsed cerebral blood velocity, it would need roughly 65% increase to restore it to the calculated healthy baseline v_{cereb} for normal tension. Overdosing patient would only push patient's heart rate lower to safety limits.

Appendix B: Model Outputs for implementation

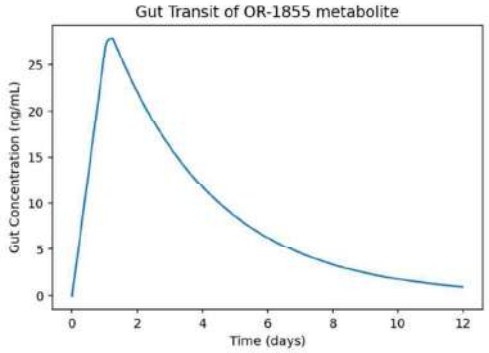


Figure 13: Gut (Time Delay) concentration over time

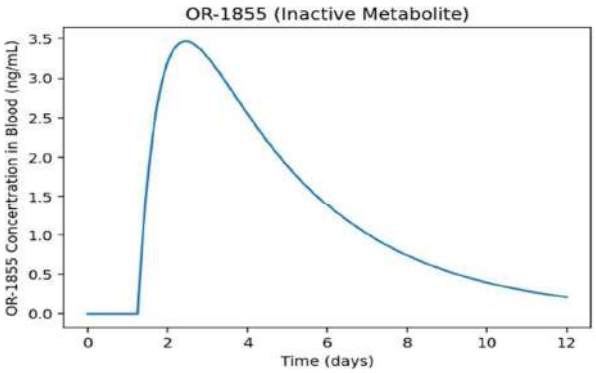


Figure 14: Blood concentration of OR-1855 metabolite over time

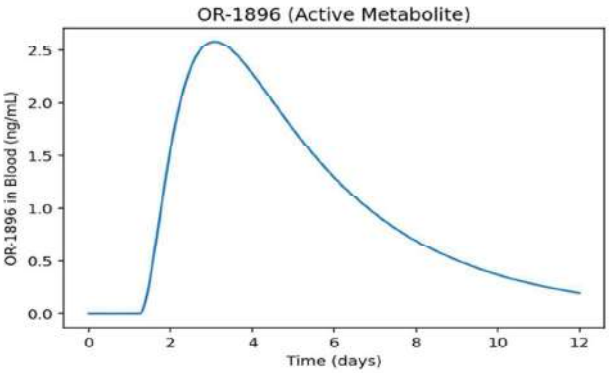


Figure 15: Blood concentration of OR-1896 metabolite over time

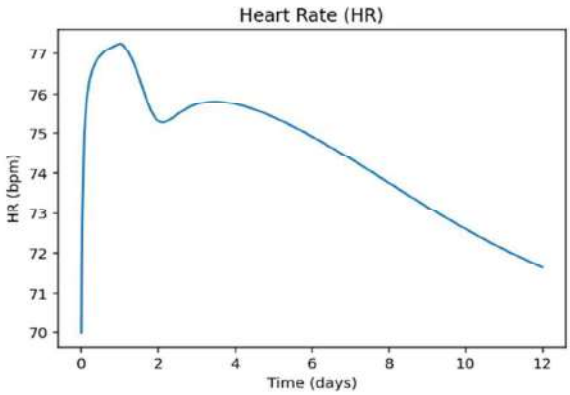


Figure 16: Heart Rate (BPM) change over time

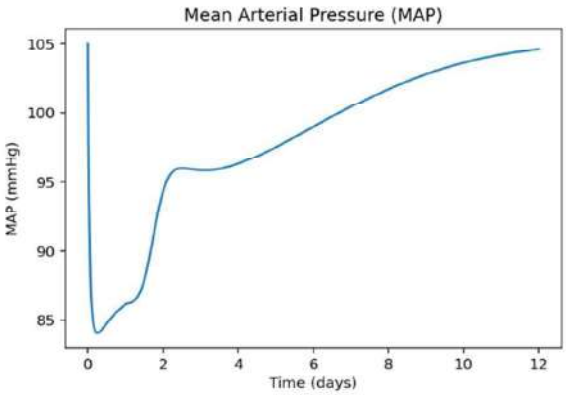


Figure 17: MAP change over time

Appendix C: Parameterisation additional material

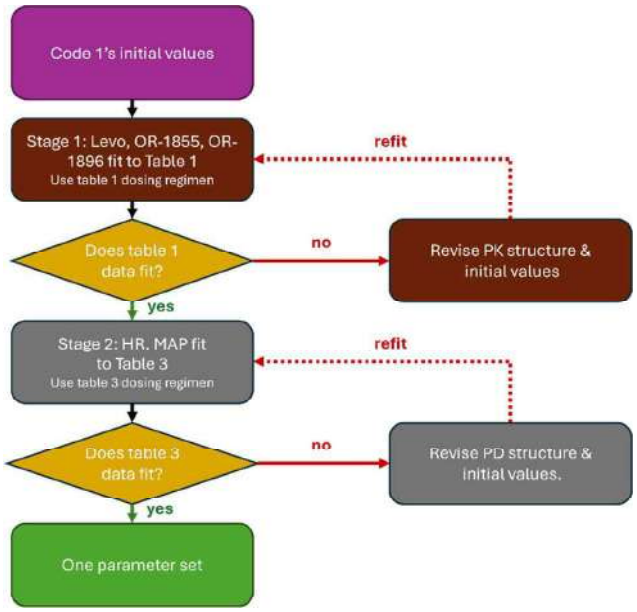


Figure 18: Parameterisation flow chart. Based from (Mann & Allenby, 2026)

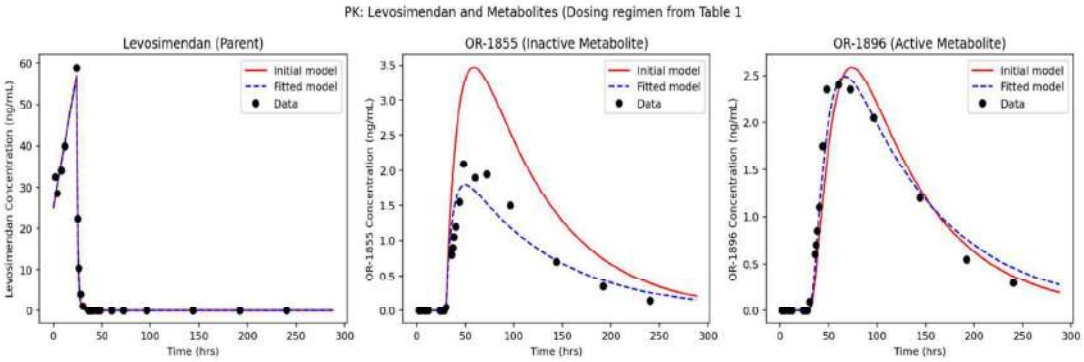


Figure 19: Fitted Table 1 data with original and parameterised model

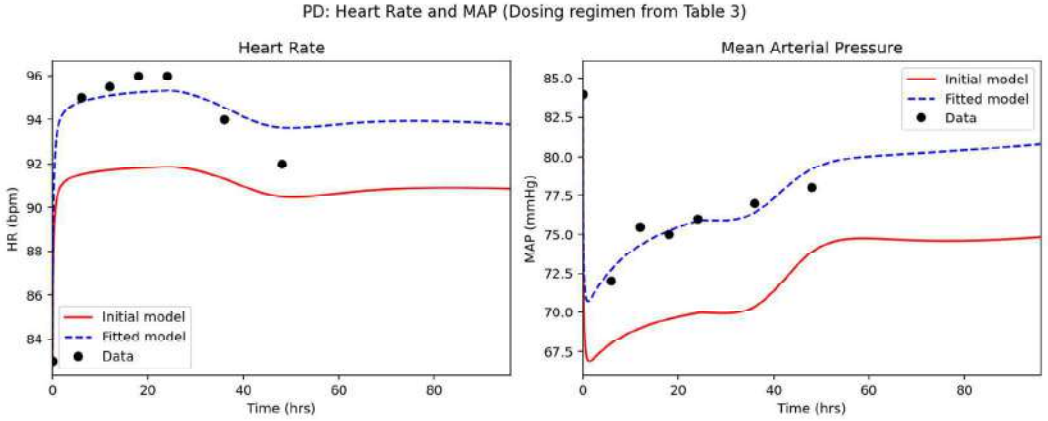


Figure 20: Fitted Table 3 data with original and parameterised model

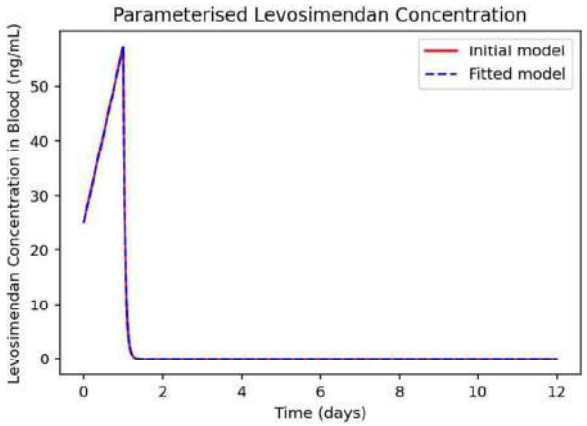


Figure 21: Parameterised blood Levosimendan concentration over time

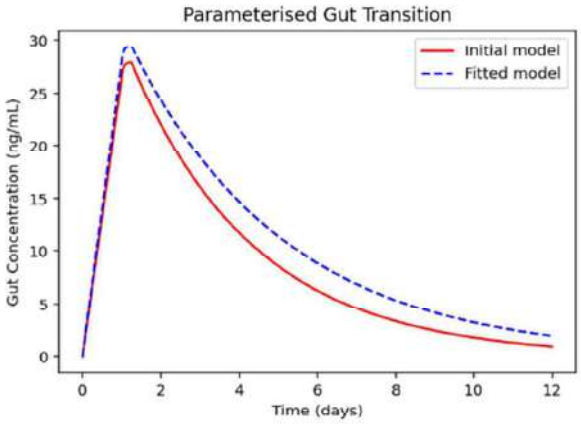


Figure 22: Parameterised gut (time delay) concentration over time

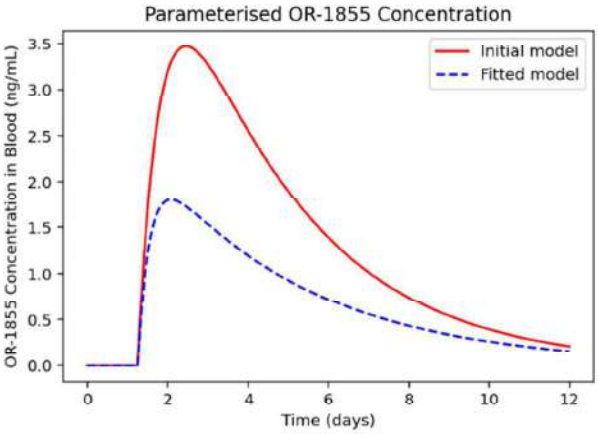


Figure 23: Parameterised blood OR-1855 concentration over time

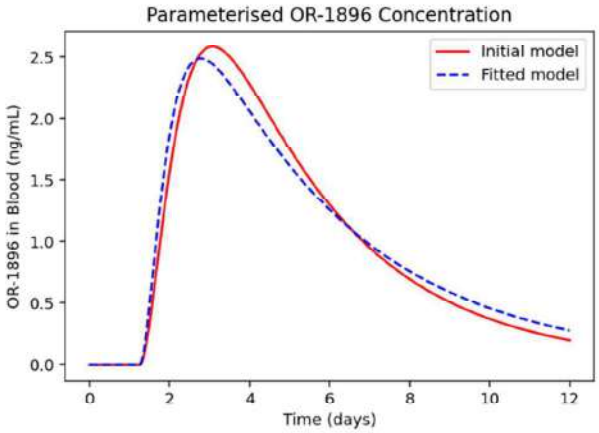


Figure 24: Parameterised blood OR-1896 concentration over time

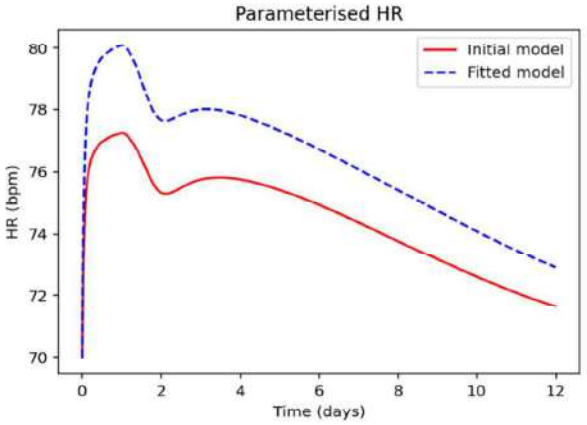


Figure 25: Parameterised heart rate change over time

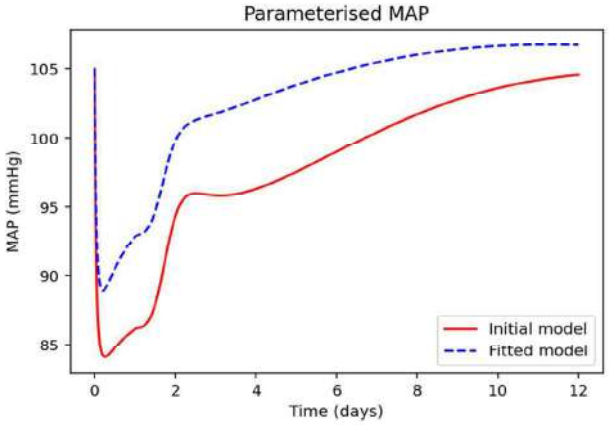


Figure 26: Parameterised MAP change over time

Appendix D: Sobol Sensitivity Analysis additional material

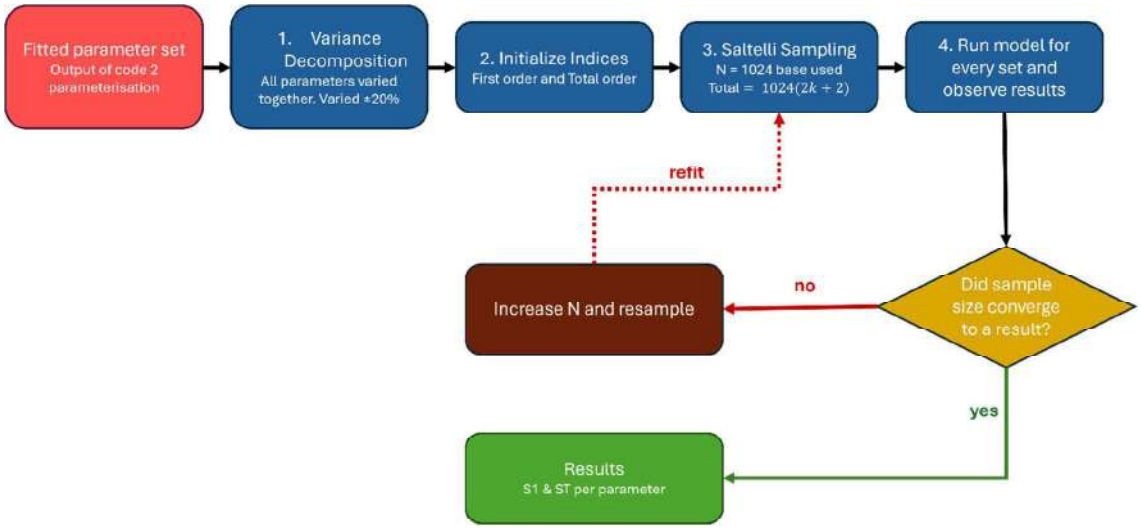


Figure 27: Flow chart of Sobol Sensitivity Analysis. Adapted from (Mann & Allenby, 2026)

v_cereb peak (cm/s)

Parameter	S1	S1 conf	ST	ST conf
gut_rate	0.012	0.011	0.016	0.002
Vm_ratio	0.021	0.013	0.031	0.003
ac_rate	0.000	0.001	0.000	0.000
m2_half	0.016	0.011	0.024	0.003
f0	0.012	0.012	0.025	0.003
keo_thalf	0.006	0.008	0.009	0.001
Emax_R	0.045	0.020	0.046	0.004
EC50	0.064	0.023	0.068	0.006
dSVmax	0.546	0.059	0.547	0.047
Emax_HR	0.258	0.040	0.262	0.022
k_HR	0.001	0.001	0.000	0.000
keo_HR_thalf	-0.000	0.000	0.000	0.000

v_cereb day 7 (cm/s)

Parameter	S1	S1 conf	ST	ST conf
gut_rate	0.001	0.002	0.001	0.000
Vm_ratio	0.133	0.028	0.136	0.013
ac_rate	0.001	0.002	0.001	0.000
m2_half	0.169	0.035	0.172	0.017
f0	-0.000	0.000	0.000	0.000
keo_thalf	0.004	0.005	0.003	0.000
Emax_R	0.012	0.010	0.013	0.001
EC50	0.134	0.031	0.136	0.012
dSVmax	0.413	0.050	0.415	0.035
Emax_HR	0.128	0.030	0.130	0.011
k_HR	-0.000	0.000	0.000	0.000
keo_HR_thalf	-0.000	0.001	0.000	0.000

Figure 28: Parameter driven v_cereb results from Sobol Analysis

Appendix E: Additional plots for dose-specific patient treatments

Note: Some plots used in report already. This is the full Code #3 output

Hypertensive Group

S0175: Stroke (Baseline)

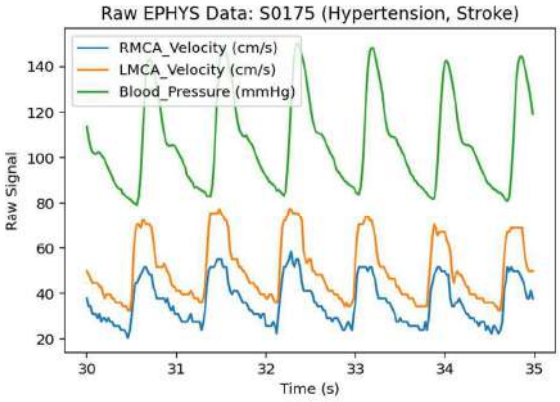


Figure 29: Raw patient data (S0175)

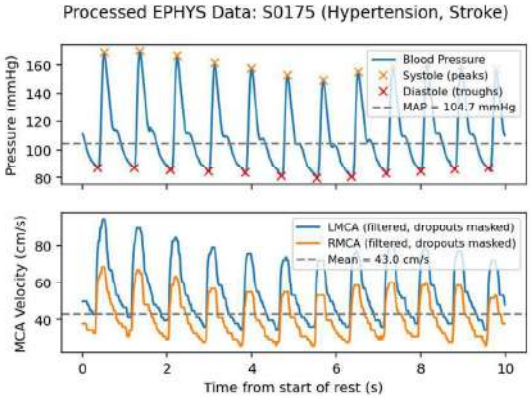


Figure 30: EPHYS processed patient data (S0175)

S0157: No Stroke (Target)

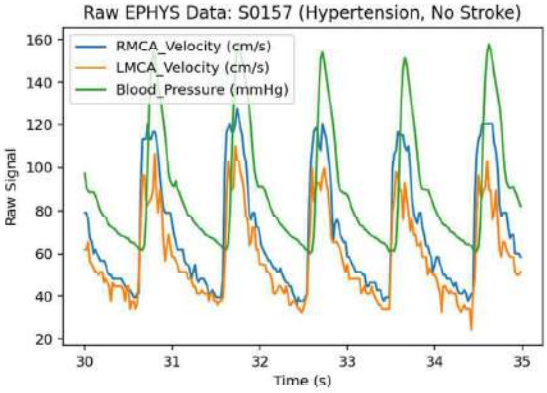


Figure 31: Raw patient data (S0157)

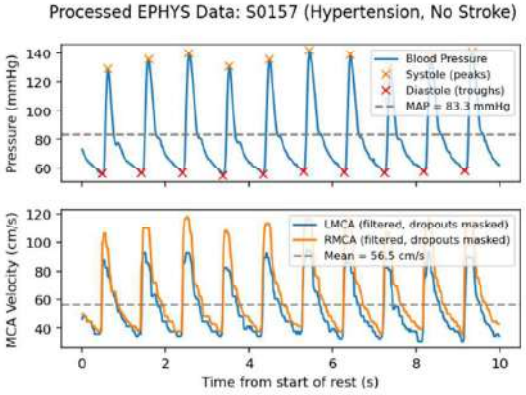


Figure 32: EPHYS processed patient data (S0157)

Normotensive Group

S0199: Stroke (Baseline)

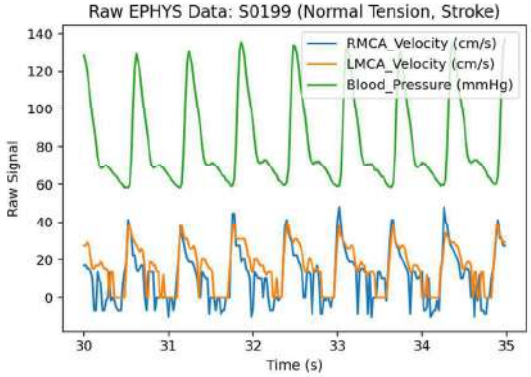


Figure 33: Raw patient data (S0199)

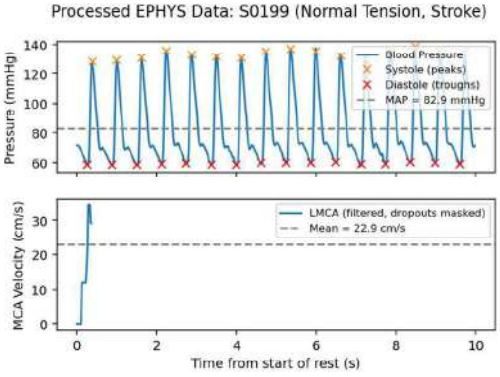


Figure 34: EPHYS processed patient data (S0199)

S0200: No Stroke (Target)

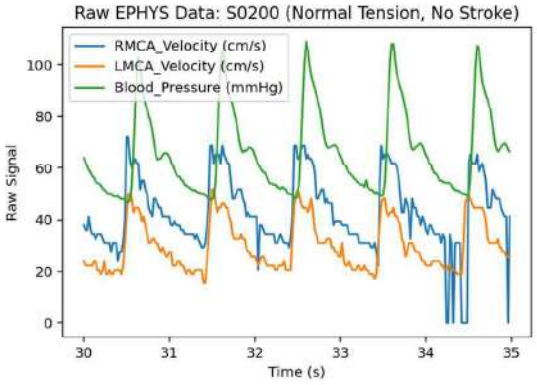


Figure 35: Raw patient data (S0200)

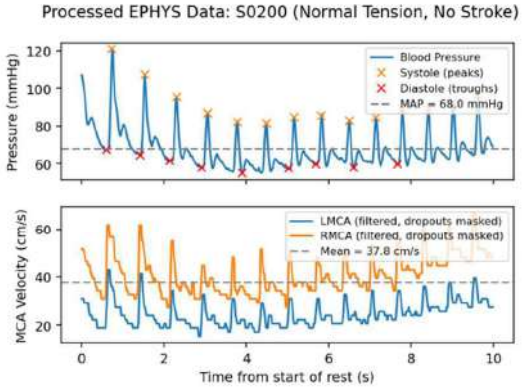


Figure 36: EPHYS processed patient data (S0200)

Table 1 Dosing Regimen: 0.1ug/kg/min x 24 hrs, every 5 days

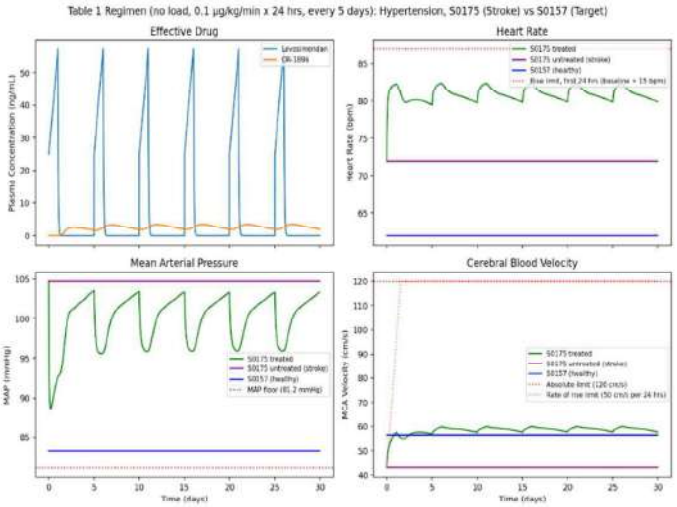


Figure 37: Table 1 dosing regimen for hypertension group

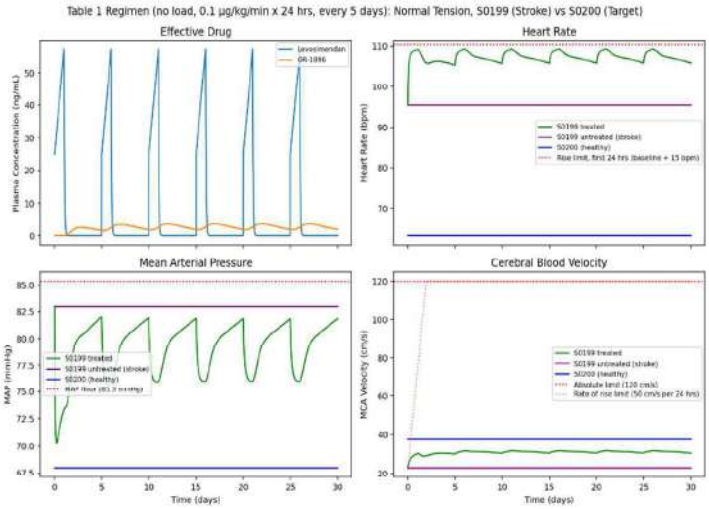


Figure 38: Table 1 dosing regimen for normal tension group

Table 3 Dosing Regimen: 12ug/kg load + 0.2ug/kg/min x 24 hrs, every 5 days

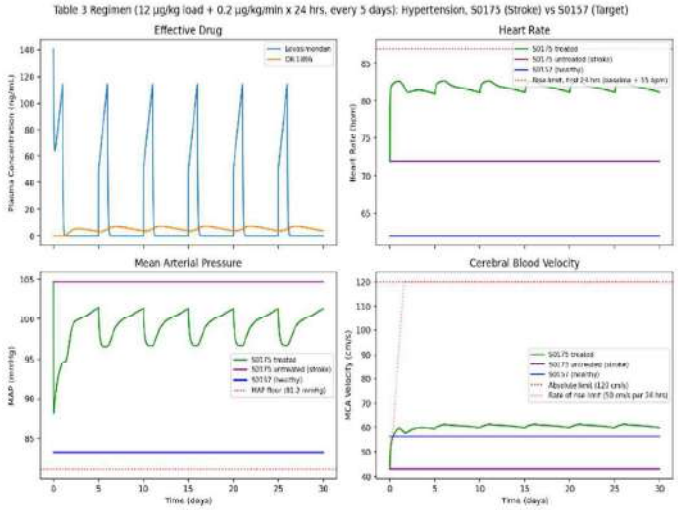


Figure 39: Table 3 dosing regimen for hypertension group

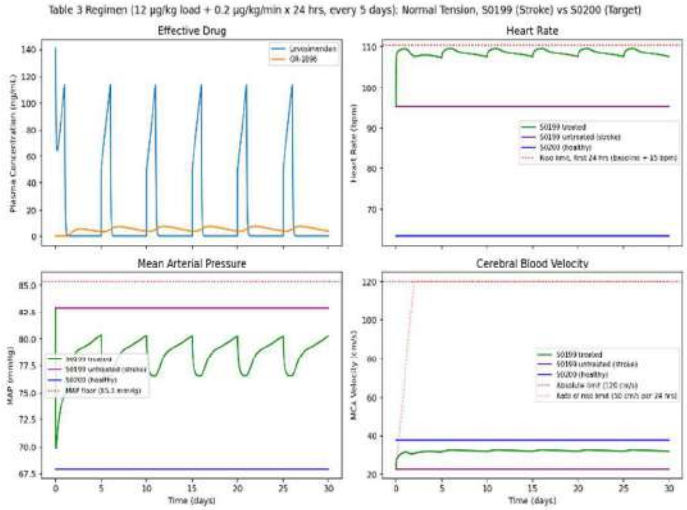


Figure 40: Table 3 dosing regimen for normal tension group

Test 1 Dosing Regimen: 0.07ug/kg/min x 24 hrs, every 5 days

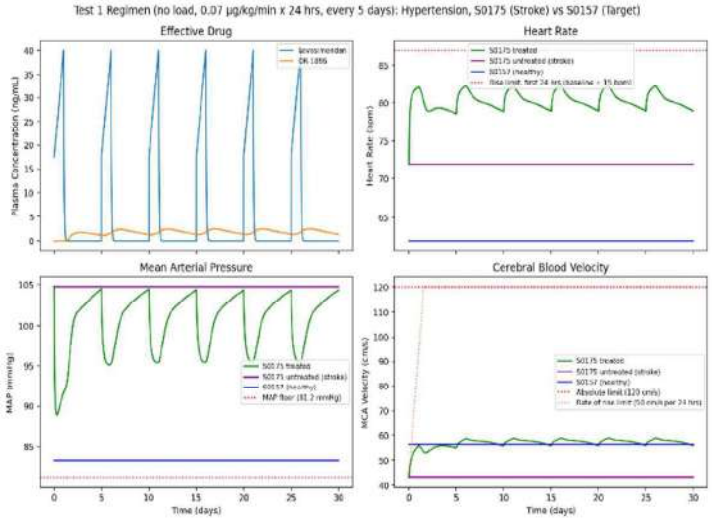


Figure 41: Dosing Test 1 regimen for hypertension group

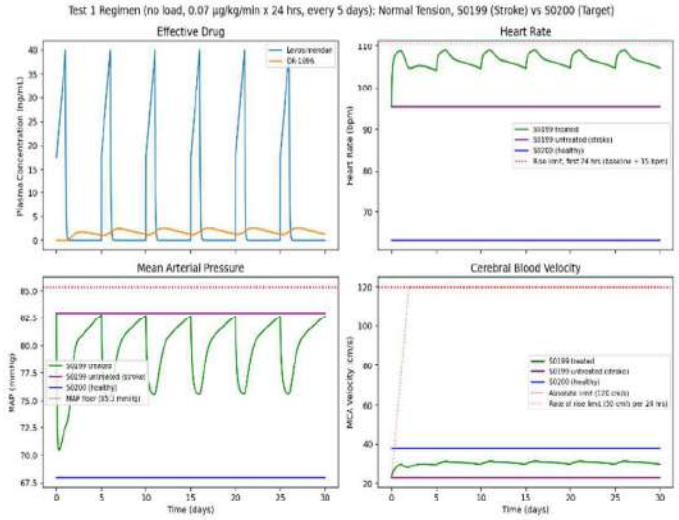


Figure 42: Dosing Test 1 regimen for normal tension group

Test 2 Dosing Regimen: 15ug/kg load + 0.3ug/kg/min x 24 hrs, every 5 days

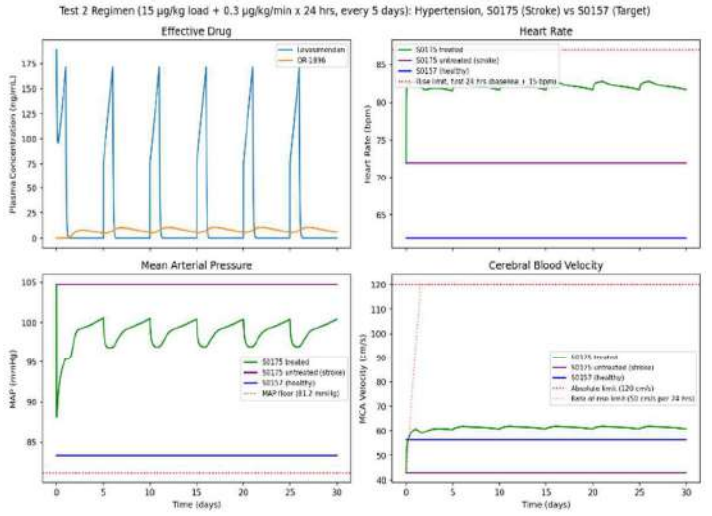


Figure 43: Dosing Test 2 regimen for hypertension group

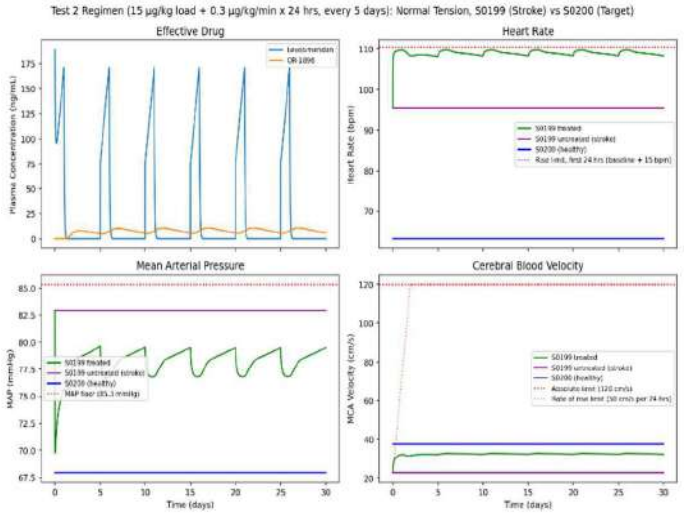


Figure 44: Dosing Test 2 regimen for normal tension group